MID-TERM EXAMINATION

(Course Name: B.Tech IT/ CSE/ CSE-AI/ AI&ML) (Semester: 3rd)

(October, 2023) OFF LINE mode

Subject Code: BCS 203	**Subject: Discrete Structures**
Time : 1 ½Hours	**Maximum Marks : 30**
Note: Q. 1 is compulsory.	

Q1		**(2.5*4)**	
	(a) What is the logical translation of the following statement? i) "None of my friends are perfect." The following notations are used: F(x): x is a friend, and P(x): x is perfect. ii) "Gold and silver ornaments are precious". The following notations are used: G(x): x is a gold ornament S(x): x is a silver ornament P(x): x is precious		
	(b) Which of the following logical expressions are equivalent I. $P \vee \sim Q$ II. $\sim (\sim P \wedge Q)$ III. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$ IV. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge Q)$		
	(c) Draw the Hasse Diagram of a set S = {1, 2, 3, 4, 6, 8, 12, 24} over the division operation.		
	(d) A bag contains 10 red balls, 10 white balls, and 10 blue balls. What is the minimum no. of balls you have to choose randomly from the bag to ensure that we get 4 balls of same color?		
Q2	**(Attempt any Two Parts) UNIT-1**	**(5,5)**	
	(a) **For all $n \geq 1$, prove that, $1.2 + 2.3 + 3.4 + \ldots + n(n+1) = \{n(n+1)(n+2)\} / 3$, by using mathematical induction.**		
	(b) *If P(x) and Q(x) be a proposition, then Identify whether a given formula is valid/invalid.* i) $\forall x\,(P(x) \rightarrow Q(x)) \rightarrow \forall x\,(P(x)) \rightarrow \forall (Q(x))$ ii) $\forall x\,(P(x) \wedge Q(x)) \leftrightarrow \forall x\,(P(x)) \wedge \forall (Q(x))$		
	(c) Consider the following two statements. *SS*1: If a candidate is known to be corrupt, then he will not be elected *SS*2: If a candidate is kind, he will be elected Which of the following statements is/are follows from *SS*1 and *SS*2 as per sound inference rules of logic? (i) If a person is known to be corrupt, he is kind (ii) If a person is not known to be corrupt, he is not kind (iii) If a person is kind, he is not known to be corrupt (iv) If a person is not kind, he is not known to be corrupt		
Q3	**(Attempt any Two Parts) UNIT-2**	**(5,5)**	
	(a) Prove that the rational number set Q is countable.		
	(b) Determine the set of positive integers between 1 and 1000 which are divisible by 2, 3, and 5. OR		
	(c) Show whether the relation (x, y) ∈ R, if, x ≥ y defined on the set of positive integers is a partial order relation.		